

BILKENT UNIVERSITY

Engineering Mathematics I Midterm 1					
Code : <i>Math 241</i> Acad. Year: <i>2013-2014</i> Semester : <i>Spring</i> Date : <i>07.04.2014</i> Time : <i>17:40</i> Duration : <i>120 minutes</i>			Last Name: Name : Section: : Student No.: Signature:		
			4 QUESTIONS ON 4 PAGES TOTAL 100 POINTS		
1	2	3	4		

No partial credit will be given for unsupported answers!

1. (25 pts) "Two square matrices A and B are called similar if there exists an invertible matrix P such that $A = P^{-1}BP$."

Show that if A and B are similar, then $\det A = \det B$

$$\begin{aligned}
 A &= P^{-1}BP \Rightarrow |A| = |P^{-1}BP| \\
 &= |P^{-1}| |B| |P| \\
 &= \frac{1}{|P|} \cdot |B| \cdot |P| \\
 &= \frac{1}{\cancel{|P|}} \cdot \cancel{|P|} \cdot |B| \\
 &\Rightarrow |A| = |B|
 \end{aligned}$$

2. (25 pts) Consider the system

$$x - 2y - z + 2t = 0$$

$$-x + 2y + 2z - t = 2$$

$$x + y + z + t = 4$$

$$x + y + 2z + 2t = 6$$

(a) Solve the system if it is consistent.

(b) Find the rank of the coefficient matrix.

$$a) \left[\begin{array}{cccc|c} 1 & -2 & -1 & 2 & 0 \\ -1 & 2 & 2 & -1 & 2 \\ 1 & 1 & 1 & 1 & 4 \\ 1 & 1 & 2 & 2 & 6 \end{array} \right] \xrightarrow{\text{e.r.o.}} \left[\begin{array}{cccc|c} 1 & -2 & -1 & 2 & 0 \\ 0 & 4 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

the system is consistent.

nb unknowns > nb of nonzero eqns

$\Rightarrow$ infinitely many solns.

x, y, z : leading variables } let $t = \alpha \Rightarrow \begin{aligned} z &= 2 - \alpha \\ y &= \alpha \\ x &= 2 - \alpha \end{aligned}$
 t : free variable

then, $\begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$ is the general soln.

$$b) \left[\begin{array}{cccc} 1 & -2 & -1 & 2 \\ -1 & 2 & 2 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 2 \end{array} \right] \xrightarrow{\text{e.r.o.}} \left[\begin{array}{cccc} 1 & -2 & -1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

As the nb of leading variables is 3, the rank of the coef. matrix is 3.

3. (25 pts) Show that the set $S = \{(1, 2, -1), (-3, 1, 0), (2, -3, 2)\}$ is a basis for $\mathbb{R}^3$

S is linearly independent:

$$\text{Let } c_1(1, 2, -1) + c_2(-3, 1, 0) + c_3(2, -3, 2) = (0, 0, 0)$$

$$\text{Then } \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 2 & 0 \\ 2 & 1 & -3 & 0 \\ 1 & 0 & 2 & 0 \end{array} \right] \xrightarrow{\text{e.r.o.}} \left[\begin{array}{ccc|c} 1 & -3 & 2 & 0 \\ 0 & 4 & -1 & 0 \\ 0 & 0 & 7 & 0 \end{array} \right]$$

system has the
unique trivial
soln.

$\Rightarrow c_1 = c_2 = c_3 = 0 \Rightarrow S$ is linearly independent.

S spans $\mathbb{R}^3$

As we know that the $\dim \mathbb{R}^3 = 3$ and S is a linearly indep. set of 3 elts, it spans $\mathbb{R}^3$.

$\Rightarrow S$ is a basis for $\mathbb{R}^3$.

4. (25 pts) "A nonempty subset W of a vector space V is called subspace of V iff W is closed under addition and scalar multiplication"

Show that the set $W = \{(x_1, x_2) | ax_1 + bx_2 = 0\}$ is a subspace of $\mathbb{R}^2$

As $a \cdot 0 + b \cdot 0 = 0$, $(0, 0) \in W \Rightarrow W \neq \emptyset$

Hence we can use the written thm.

(i) $\forall u = (u_1, u_2) \in W \Rightarrow au_1 + bu_2 = 0$

$\forall v = (v_1, v_2) \in W \Rightarrow av_1 + bv_2 = 0$

Then $u+v = (u_1, u_2) + (v_1, v_2) = (u_1+v_1, u_2+v_2)$

And $a(u_1+v_1) + b(u_2+v_2) = au_1 + av_1 + bu_2 + bv_2$
 $= \underbrace{au_1 + bu_2}_0 + \underbrace{av_1 + bv_2}_0$

$\Rightarrow u+v \in W$ (W is closed under addition)

(ii) $\forall u = (u_1, u_2) \in W \Rightarrow au_1 + bu_2 = 0$
 $\forall c \in \mathbb{R}$

$c \cdot u = c(u_1, u_2) = (cu_1, cu_2)$

And $a(cu_1) + b(cu_2) = c(au_1) + c(bu_2)$
 $= c(\underbrace{au_1 + bu_2}_0) = 0 //$

$\Rightarrow c \cdot u \in W$ (W is closed under scalar mult.)

$\Rightarrow W$ is a subspace of $\mathbb{R}^2$